

Shubhadeep Datta

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Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (AI/ML) | CGPA: 7.65 – Expected 2027

Andhra Pradesh

Bhavan's Tripura Vidya Mandir

CBSE Class X & Class XII

Sep 2023 – Present

Tripura

2019 – 2021

Experience

Research Intern, School of Cybersecurity & Digital Forensics

2026

National Forensic Sciences University (NFSU), Tripura Campus

Under Dr. Priya Saha

- Engineered a Federated Learning pipeline from scratch – FedAvg aggregation, non-IID partitioning across 5 clients – with zero dependency on external FL frameworks
- Achieved **88.36% peak accuracy** benchmarking 7 normalisation strategies over 20 communication rounds; identified GroupNorm as the optimal accuracy-fairness trade-off
- Built an automated test harness looping training across all configurations and auto-generating comparison plots/metrics – replacing a manual, multi-day process with a single run

Leadership & Extracurricular Activities

Marketing Lead

2024 – 2025

Microsoft Student Chapter, VIT-AP University

- Grew event participation **30%** as point of contact for 100+ students across 3+ ML workshops, coordinating outreach with a team of 15+ volunteers

Vice President

2025

Northeast Cultural Club, VIT-AP University

- Founded and scaled the club to 40+ members, reaching 150+ students campus-wide

Projects

fix-protocol-mcp – FIX Trading Protocol MCP Server | Python, MCP (FastMCP), Docker, pytest

2026

- Designed and built a protocol-compliant FIX 4.4 server – the standard messaging protocol for order entry and execution reporting in trading systems – with four tools to parse, validate, build, and explain messages against a bundled field dictionary, zero external API dependency
- Achieved a **100% pass rate across 34 automated pytest unit/integration tests** validating protocol compliance in a Dockerized, CI-ready build; merged into the community-maintained awesome-mcp-servers directory after passing its automated build/introspection verification (Glama)

BaitBlock – Clickbait Detection | Python, Node.js, NLP, SBERT

2025

- Authored and maintained an automated pytest regression suite covering the scoring function and REST endpoints, achieving **50%+ test coverage** across a full-stack detector with dual interchangeable backends (Node.js heuristic + Python SBERT/spaCy) behind one REST API; changes reviewed via GitHub pull requests
- Engineered defense-in-depth SSRF protection across a 4-tier article acquisition pipeline (HTTP -> AMP/RSS -> headless -> Wayback Machine), validated across every redirect hop

Typical – Gaze-Verified Typing Trainer | Next.js, TypeScript, MediaPipe, PostgreSQL

2026

- Integrated MediaPipe's FaceLandmarker model into an on-device gaze-integrity engine verifying typing focus via calibrated pitch/blendshape baselines, running in a Web Worker with zero video leaving the browser
- Architected a two-tier Next.js 16 app (IndexedDB guest mode -> PostgreSQL/Prisma accounts) with dual-layer anti-cheat combining client-side gaze verification and server-side keystroke-timing bot detection

CareLink | React Native, Expo, TypeScript, Google Gemini API

2025

- Developed a cross-platform caregiving app (6 screens, 11 components) with medication adherence tracking, live GPS fitness monitoring, and a Gemini 1.5 Flash AI health assistant
- Won 1st Prize, Healthcare for the Elderly category** at an App Development Hackathon among 50+ teams

LeafScan – Pest Detection & Recommendation | Python, PyTorch, Streamlit, Docker

2025

- Achieved **0.761 Macro-F1** and **95% Top-3 accuracy** training a temperature-calibrated ResNet-50 across 9 pest species, with Grad-CAM explainability; reduced calibration error **45%**

Technical Skills

Testing & QA: Unit/Functional/Integration Testing, pytest, Test Automation, Code Coverage Analysis, REST API Testing, Protocol Testing (FIX)

Languages: Java, Python, JavaScript, TypeScript, C/C++, SQL

Backend: REST API, MVC, Authentication, Spring Boot

Cloud & DevOps: Git, GitHub, CI/CD, Docker, AWS, Azure

AI / Machine Learning: TensorFlow, PyTorch, CNN, NLP, Federated Learning